

Gated Recurrent Units (GRU): Comprehensive Notes

1. Introduction

The Gated Recurrent Unit (GRU) is a sophisticated architecture in the family of Recurrent Neural Networks (RNNs). Introduced in 2014 by Kyunghyun Cho et al., it was designed to address the challenges of vanishing gradients in standard RNNs while remaining computationally more efficient than the Long Short-Term Memory (LSTM) networks.

2. Architectural Components

Unlike standard RNNs, GRUs utilize a gating mechanism to control the flow of information across time steps. They consist of two primary gates:

- **Update Gate (z_t):** Determines how much of the past hidden state to carry forward to the current state.
- **Reset Gate (r_t):** Determines how much of the past information to "forget" or ignore when computing the candidate hidden state.

3. Mathematical Formulation

At each time step t , given input x_t and previous hidden state h_{t-1} :

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

$$\tilde{h}_t = \tanh(W \cdot [r_t \odot h_{t-1}, x_t])$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

4. Applications in Text Generation

GRUs are highly effective for sequential data modeling. By maintaining a compressed memory of previous tokens in the sequence, they can predict subsequent words with high contextual awareness. Their reduced parameter set compared to LSTMs makes them a preferred choice for rapid prototyping and smaller-scale text generation projects.